Bleeding following Trauma

This guideline is for patients with suspected or confirmed internal bleeding, or clinically significant external bleeding, following blunt or penetrating trauma.

- Immediately control life-threatening external bleeding:
  - Apply manual external pressure.
  - Apply an arterial tourniquet if life-threatening bleeding from a limb is not easily controlled by manual external pressure.
  - Consider packing the wound if arterial tourniquet application is not indicated, using a haemostatic dressing if one is available.
- Apply a chest seal device to penetrating chest injuries or a standard dressing if a chest seal device is not available.
- Apply a pelvic splint if the mechanism of injury could have caused a fractured pelvis.
- Administer 0.9% sodium chloride IV (using a fluid warming device if available), if the patient has signs of hypovolaemia:
  - 500 ml IV for an adult.
  - 10 ml/kg IV for a child.
  - Repeat as required, minimising the total volume.
- Administer tranexamic acid IV:
  - 2 g IV for an adult.
  - See the paediatric drug dose tables for a child.
- If shock is severe:
  - Expedite transport.
  - Call for blood, provided there is an established protocol for this in the area, particularly if extrication will be delayed and/or transport will be prolonged.
  - Consider seeking clinical advice and commencing a metaraminol infusion in an adult, if transport time is prolonged and the systolic BP is less than 80 mmHg systolic. Titrate the infusion rate to maintain a systolic BP of 80-90 mmHg.
- Provide active external warming whenever feasible.
- Splint long bone fractures.
- Notify receiving hospital personnel early. Activate a critical haemorrhage code if indicated, provided there is an established protocol for activation by ambulance personnel in the area.

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Backup

- Backup from a Paramedic must be requested if the patient has signs of hypovolaemia.
- Backup from an ICP/CCP must be requested if the patient has severe shock.
- Backup must not delay rapid transport to an appropriate hospital if shock is severe.

Referral and transport

- Expedite transport, particularly when there are signs of severe shock. This is particularly important with penetrating truncal trauma when the transport time to an appropriate hospital is short.
- Transport the patient direct to a major trauma hospital whenever this is feasible and safe.
- Transport the patient direct to a major trauma hospital with vascular surgical facilities if there are clear signs of arterial bleeding from a large artery, for example the brachial, popliteal or femoral artery. Clinical judgement must be used if a major trauma hospital with vascular surgical facilities is significantly further away than the closest major trauma hospital, and personnel should have a low threshold for seeking clinical advice in this setting.

Additional information

Arterial tourniquet application

- Applying a tourniquet is a balance of risk.
  - Tourniquets help control severe bleeding but can cause significant tissue ischaemia and damage (for example to nerves) and should only be used when external pressure is insufficient to control life-threatening bleeding.
  - Applying a tourniquet will cause ischaemia of the limb distal to the tourniquet and this may compromise the success of surgical reattachment.
- When applying a tourniquet:
  - Remove clothing from the limb if possible.
  - Apply as distally as possible.
  - Do not apply over a joint.
  - Tighten until bleeding clearly stops. The tourniquet must be tight enough to stop arterial flow.
  - Leave the wound exposed so that it can be observed for bleeding.
  - Record the time of application.

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- Reassess the tourniquet frequently following treatment. It may need to be further tightened if blood pressure improves.
- If the tourniquet is applied to a forearm or lower leg, the presence of two bones may limit the pressure that can be applied to vessels. If bleeding continues despite the tourniquet being tightened maximally, place the tourniquet on the upper arm or thigh.
- The tourniquet may need to be very tight if applied over a large thigh. In this setting a second tourniquet may be required proximal to the first tourniquet.
- Re-evaluate the need for a tourniquet after bleeding is controlled, IV access has been obtained and appropriate fluid resuscitation has been commenced:
  - If the source of the bleeding is a single lacerated artery, for example a lacerated brachial or radial artery, it is preferable that bleeding is controlled with direct pressure and then the tourniquet is released, particularly when transport time to hospital is over 30 minutes.
  - If bleeding cannot be controlled with direct pressure the tourniquet should be reapplied.
- A tourniquet may be used to provide pressure over a focal dressing, for example directly over a lacerated brachial artery. In this setting the tourniquet needs to be tight enough to control bleeding and not necessarily tight enough to stop arterial flow.

Pelvic splinting

- It is often difficult to determine accurately that a pelvis is fractured by clinical examination. Assume that the pelvis is fractured if the patient has pain in the pelvic area or is unable to report pain and has a suitable mechanism of injury.
- Remove clothing prior to placing a splint whenever feasible.
- Firmly splint the pelvis using a service supplied dedicated pelvic splint.
- Do not use a sheet to splint the pelvis because a sheet cannot be applied firmly enough.

Penetrating chest wounds and chest seal devices

- Use a service supplied dedicated chest seal provided one is available.
- Do not seal the wound with other devices (such as a defibrillator pad) as this risks the development of tension pneumothorax.
- A chest seal device is not required if the patient has been intubated and ventilated, as positive pressure in the chest from ventilation will prevent air entry into the chest via the wound.

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- Do not attempt to create a sealed dressing on three sides to form a valve, as this is rarely effective.

Administration of 0.9% sodium chloride

- 0.9% sodium chloride IV should be titrated to clinical signs of intravascular volume and perfusion.
- Large volumes of crystalloid may worsen bleeding by contributing to coagulopathy, particularly in the context of severe and uncontrolled bleeding:
  - This is why the total volume of 0.9% sodium chloride should be minimised.
  - Clinical judgment is required when determining the volume administered, but it should be uncommon to exceed one litre in an adult or 20 ml/kg in a child, and it should be rare to exceed two litres in an adult or 40 ml/kg in a child.
  - Have a low threshold for seeking clinical advice if shock is severe and transport time to the most appropriate hospital is prolonged.
- Ambient temperature IV fluid can cause hypothermia which worsens bleeding by contributing to coagulopathy. IV fluid should be warmed using a dedicated warming device if one is available.
- Do not warm IV fluid in microwaves because this can result in unreliable fluid temperatures and/or may damage the bag.

Tranexamic acid (TXA)

- TXA reduces bleeding by reducing clot break down via fibrinolysis, and this is associated with reduced mortality rates in bleeding secondary to trauma.
- TXA is indicated if there are signs of clinically significant bleeding or signs of hypovolaemia.
- TXA administration is not a priority and it should usually be administered with the first dose of 0.9% sodium chloride.

Blood

- Requests for blood to be delivered to the scene must not compromise rapid transport to the most appropriate hospital.
- Blood should only be requested if:
  - Shock is severe, and
  - There is an established protocol in the area for blood to be delivered to the scene, and

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- Blood can be administered 15 or more minutes before the patient can arrive in the most appropriate hospital.

- Blood may be available via staging at a small hospital/medical facility en route to the most appropriate hospital. In this setting personnel must have a low threshold for seeking advice from the on-call doctor for the service.

- ICPs/CCPs may continue blood administration initiated by a medical practitioner following their verbal instructions but should have a low threshold for seeking clinical advice.

- Requests for blood to be delivered to the scene outside of an established protocol must be made via the on-call doctor for the service. In this setting:
  - It will usually take approximately 15 minutes to arrange for blood to be prepared, plus transport time to the scene. This means that for such a request to be appropriate, the patient should usually have severe shock and a prolonged transport time.
  - Personnel cannot administer blood without specific instructions from the on-call doctor for the service.

Patient temperature

- It is important to keep the patient warm because hypothermia worsens bleeding by contributing to coagulopathy.

- Passive warming (for example with external blankets) is rarely effective, and the patient should also be actively warmed.

- It is important to document the patient's temperature.

Bleeding from wounds

- The best method of controlling severe external bleeding is usually by providing firm and sustained manual pressure using a dressing directly over the site of the bleeding, before applying a focal dressing and a pressure bandage.

- A haemostatic dressing (if available) will reduce bleeding, but do not delay manual external pressure while obtaining a haemostatic dressing.

- Topical adrenaline is not useful in the setting of severe external bleeding. Topical adrenaline may be considered in the setting of non-severe bleeding if a haemostatic dressing is unavailable.
  - Dilute adrenaline to 1:10,000 using 0.9% of sodium chloride. For example, dilute 2 mg of adrenaline to a total of 20 ml.
  - Soak a dressing in 1:10,000 adrenaline. Place the dressing into the wound or over the bleeding site, and continue to provide pressure over the wound.

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Critical haemorrhage codes and the assessment of blood consumption score

- Critical haemorrhage codes such as Code Crimson and Code Red are used in many major trauma hospitals to accelerate in-hospital clinical interventions to stop life-threatening bleeding.
- Notify ED personnel as soon as possible if shock is severe, to enable early activation of a critical haemorrhage code. Where an agreed protocol is in place this may be activated by transporting personnel.
- The criteria for critical haemorrhage code activation are:
  - An assessment of blood consumption (ABC) score greater than or equal to 2, or
  - The patient has received blood products.
- The ABC score is composed of (score one for each):
  - Heart rate greater than or equal to 120/min.
  - Systolic BP less than or equal to 90 mmHg.
  - Penetrating truncal trauma.
  - A positive ultrasound scan for free fluid in the abdomen, pelvis, or chest.

Penetrating objects

- Penetrating objects should usually be removed from a limb, as this will allow direct pressure to be more reliably applied. Clinical judgement should be used if the object is large (for example a knife rather than a nail) and appears to be near an artery.
- Do not remove penetrating objects from the head or truncal area because severe bleeding may occur that cannot be compressed.
- Provide manual pressure directly around the object if severe bleeding is present and the object is not removed.

Metaraminol

- Metaraminol is not a priority in patients with haemorrhage. However, as shock progresses vasodilation will often develop, and it is important to minimise the total volume of IV fluid administered.
- Metaraminol should only be administered to adult patients who continue to have severe hypotension following IV fluid administration.
- If a metaraminol infusion has been commenced, this should be accompanied by further fluid boluses if indicated.

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- The aim of administering metaraminol is not to restore normal perfusion, but to maintain sufficient perfusion of vital organs without increasing blood loss.

More Information

Version history

28 October 2024 1.0.5

New: Guideline released.